

Exercise JA

Q1. The total number of ways in which 5 balls of different colours can be distributed among 3 persons so that each person gets at least one ball is :

- (A) 75 (B) 150 (C) 210 (D) 243

Ans. B

Sol. We know that total number of ways of distributing n distinct things among m persons such that each person gets at least one thing

$$= m^n - {}^m C_1 \cdot (m-1)^n + {}^m C_2 (m-2)^n - {}^m C_3 (m-3)^n + \dots$$

So the required total number of ways

$$= 3^5 - {}^3 C_1 \cdot 2^5 + {}^3 C_2 \cdot 1^5 = 243 - 96 + 3 = 150$$

Paragraph

Let a_n denote the number of all n -digit positive integers formed by the digits 0, 1 or both such that no consecutive digits in them are 0. Let b_n = the number of such n -digit integers ending with digit 1 and C_n = the number of such n -digit integers ending with digit 0.

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Q2. Which of the following is correct ?

- (A) $a_{17} = a_{16} + a_{15}$ (B) $C_{17} \neq C_{16} + C_{15}$ (C) $b_{17} \neq b_{16} + C_{16}$ (D) $a_{17} = C_{17} + b_{16}$

Ans. A

Sol. -

Paragraph

Let a_n denote the number of all n -digit positive integers formed by the digits 0, 1 or both such that no consecutive digits in them are 0. Let b_n = the number of such n -digit integers ending with digit 1 and C_n = the number of such n -digit integers ending with digit 0.

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Q3. The value of b_6 is

- (A) 7 (B) 8 (C) 9 (D) 11

Ans. B

Sol. -

Q4. Six cards and six envelopes are numbered 1, 2, 3, 4, 5, 6 and cards are to be placed in envelopes so that each envelope contains exactly one card and no card is placed in the envelope bearing the same number and moreover the card numbered 1 is always placed in envelope numbered 2. Then the number of ways it can be done is

(A) 264 (B) 265 (C) 53 (D) 67

Ans. C

Sol. -

Q5. Let $n_1 < n_2 < n_3 < n_4 < n_5$ be positive integers such that $n_1 + n_2 + n_3 + n_4 + n_5 = 20$. Then the number of such distinct arrangement $(n_1, n_2, n_3, n_4, n_5)$ is

Ans. 7

Sol. -

Q6. Let $n \geq 2$ be an integer. Take n distinct points on a circle and join each pair of points by a line segment.

Colour the line segment joining every pair of adjacent points by blue and the rest by red. If the number of red and blue line segments are equal, then the value of n is

Ans. 5

Sol. -

Q7. Let n be the number of ways in which 5 boys and 5 girls can stand in a queue in such a way that all the girls stand consecutively in the queue. Let m be the number of ways in which 5 boys and 5 girls can stand in a queue in such a way that exactly four girls stand consecutively in the queue. Then the value of $\frac{m}{n}$ is

Ans. 5

Sol. -

Q8. A debate club consists of 6 girls and 4 boys. A team of 4 members is to be selected from this club including the selection of a captain (from among these 4 members) for the team. If the team has to include at most one boy, then the number of ways of selecting the team is

(A) 380 (B) 320 (C) 260 (D) 95

Ans. A

Sol. -

Q9. Let $S = \{1, 2, 3, \dots, 9\}$. For $k = 1, 2, \dots, 5$, let N_k be the number of subsets of S each containing five elements out of which exactly k are odd. Then $N_1 + N_2 + N_3 + N_4 + N_5 =$

(A) 125 (B) 210 (C) 252 (D) 126

Ans. D

Sol. -

Q10. Words of length 10 are formed using the letters A, B, C, D, E, F, G, H, I, J. Let x be the number of such words where no letter is repeated; and let y be the number of such words where exactly one letter is repeated twice and no other letter is repeated. Then, $\frac{y}{9x} =$

Ans. 5

Sol. -

Q11. The number of 5 digit numbers which are divisible by 4, with digits from the set $\{1,2,3,4,5\}$ and the repetition of digits is allowed, is _____ .

Ans. 625

Sol. -

Q12. In a high school, a committee has to be formed from a group of 6 boys $M_1, M_2, M_3, M_4, M_5, M_6$ and 5 girls G_1, G_2, G_3, G_4, G_5 .

(i) Let α_1 be the total number of ways in which the committee can be formed such that the committee has 5 members, having exactly 3 boys and 2 girls.

(ii) Let α_2 be the total number of ways in which the committee can be formed such that the committee has at least 2 members, and having an equal number of boys and girls.

(iii) Let α_3 be the total number of ways in which the committee can be formed such that the committee has 5 members, at least 2 of them being girls.

(iv) Let α_4 be the total number of ways in which the committee can be formed such that the committee has 4 members, having at least 2 girls and such that both M_1 and G_1 are NOT in the committee together.

LIST-I	LIST-II
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P. The value of α_1 is	1. 136
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Q. The value of α_2 is	2. 189
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R. The value of α_3 is	3. 192
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S. The value of α_4 is	4. 200
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	5. 381
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	6. 461
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The correct option is:

(A) $P \rightarrow 4; Q \rightarrow 6; R \rightarrow 2; S \rightarrow 1$ (B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 3$

(C) $P \rightarrow 4; Q \rightarrow 6; R \rightarrow 5; S \rightarrow 2$ (D) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$

Ans. C

Sol. -

Q13. Five persons A,B,C,D and E are seated in a circular arrangement. If each of them is given a hat of one of the three colours red, blue and green then the number of ways of distributing the hats such that the persons seated in adjacent seats get different coloured hats is _____

Ans. 30

Sol. -

Q14. An engineer is required to visit a factory for exactly four days during the first 15 days of every month and it is mandatory that no two visits take place on consecutive days. Then the number of all possible ways in which such visits to the factory can be made by the engineer during 1-15 June 2021 is _____

Ans. 495

Sol. Out of 15 days, we have to choose four days such that no two are consecutive.

Let us reduce 4 days out of 15, so Remaining days = $15 - 4 = 11$.

Let the remaining days are $d_1, d_2, \dots, d_{11}$

$\downarrow d_1 \downarrow d_2 \downarrow d_3 \downarrow \dots \downarrow d_{11} \downarrow$

Among the 12 gaps in these 11 days, we have to pick any 4, so

$$\text{No. of ways } {}^{12}C_4 = \frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2} = 495$$

Q15. In a hotel, four rooms are available. Six persons are to be accommodated in these four rooms in such a way that each of these rooms contains at least one person and at most two persons. Then the number of all possible ways in which this can be done is _____.

Ans. 1080

Sol. The only ways in distributing 6 - persons in 4 rooms, such that each room contains at least 1 and at most 2 person is (1, 1, 2, 2).

$$\text{No. of ways of grouping} = \frac{6!}{1!1!(2!)2!2!(2!)} = 45$$

No. of ways of arranging these 4 groups in 4 rooms = $4! = 24$

So total no. of ways = $45 \times 24 = 1080$

Q16. Let $S_1 = S_1 = \{(i, j, k) : i, j, k \in \{1, 2, \dots, 10\}\}$
 $S_2 = \{(i, j) : 1 \leq i < j + 2 \leq 10, i, j \in \{1, 2, \dots, 10\}\}$
 $S_3 = \{(i, j, k, l) : 1 \leq i < j < k < l, i, j, k, l \in \{1, 2, \dots, 10\}\}$

and $S_4 = \{(i, j, k, l) : i, j, k \text{ and } l \text{ are distinct elements in } \{1, 2, \dots, 10\}\}$.

If the total number of elements in the set S_r is n_r , $r = 1, 2, 3, 4$, then which of the following statements is (are) **TRUE** ?

- (A) $n_1 = 1000$ (B) $n_2 = 44$ (C) $n_3 = 220$ (D) $\frac{n_4}{12} = 420$

Ans. A, B, D

Sol. $n_1 = 10 \times 10 \times 10 = 1000$

$$i = 1 \Rightarrow 1 \leq j \leq 8$$

$$i = 2 \Rightarrow 1 \leq j \leq 8$$

$$i = 3 \Rightarrow 2 \leq j \leq 8$$

;

;

$$i = 7 \Rightarrow j = 8$$

$$n_2 = 8 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = 44$$

$$n_3 = {}^{10}C_4 \cdot 4! = 210$$

$$n_4 = {}^{10}C_4 = 5040$$

$$\frac{n_4}{12} = 420$$

ABD

Q17. The number of 4-digit integers in the closed interval [2022, 4482] formed by using the digits 0, 2, 3, 4, 6, 7 is _____.

Ans. 569

Sol.

(1)

2	0	2	$\begin{matrix} 2, 3 \\ 4, 6, 7 \end{matrix}$
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 $\rightarrow 5$

(2)

2	0	$\begin{matrix} 3, 4 \\ 6, 7 \end{matrix}$	
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 $\rightarrow 24$

$\downarrow \quad \downarrow$
4 6

(3)

2	$\begin{matrix} 2, 3, 4 \\ 6, 7 \end{matrix}$		
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 $\rightarrow 180$

$\downarrow \quad \downarrow \quad \downarrow$
5 6 6

(4)

3			
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 $\rightarrow 216$

$\downarrow \quad \downarrow \quad \downarrow$
6 6 6

(5)

4	$\begin{matrix} 0, 2, \\ 3, 4 \end{matrix}$		
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 $\rightarrow 144$

$\downarrow \quad \downarrow \quad \downarrow$
4 6 6

Number of 4 digit integers in [2022,4482]

$$= 5 + 24 + 180 + 216 + 144 = 569$$

Q18. Consider 4 boxes, where each box contains 3 red balls and 2 blue balls. Assume that all 20 balls are distinct. In how many different ways can 10 balls be chosen from these 4 boxes so that from each box at least one red ball and one blue ball are chosen ?

- (A) 21816 (B) 85536 (C) 12096 (D) 156816

Ans. A

Sol.

3B	3B	3B	3B
2R	2R	2R	2R

$$\begin{aligned} & \begin{bmatrix} 2B & 2B & 1B & 1B \\ 1R & 1R & 1R & 1R \end{bmatrix} \times 6 \\ & \begin{bmatrix} 3B & 1B & 1B & 1B \\ 1R & 1R & 1R & 1R \end{bmatrix} \times 4 \\ & \begin{bmatrix} 1B & 1B & 1B & 1B \\ 2R & 2R & 1R & 1R \end{bmatrix} \times 6 \\ & \begin{bmatrix} 2B & 1B & 1B & 1B \\ 2R & 1R & 1R & 1R \end{bmatrix} \times 6 \end{aligned}$$

Now

$$\begin{aligned} & ({}^3C_2 \times {}^2C_1 \times {}^3C_2 \times {}^2C_1 \times {}^3C_1 \times {}^2C_1 \times {}^3C_1 \times {}^2C_1) \times 6 + \\ & ({}^3C_3 \times {}^2C_1 \times {}^3C_1 \times {}^2C_1 \times {}^3C_1 \times {}^2C_1 \times {}^3C_1 \times {}^2C_1) \times 4 \\ & + ({}^3C_1 \times {}^2C_2 \times {}^3C_1 \times {}^2C_2 \times {}^3C_1 \times {}^2C_1 \times {}^3C_1 \times {}^2C_1) \times 6 + \\ & = 6^3 [36 + 8 + 9 + 48] = 216 \times (53 + 48) = 216 \times 101 = 21816 \end{aligned}$$

Alternative :

We needed to find coeff of

$x^4 y^6, x^5 y^5$ & $x^6 y^4$ in

$$({}^3C_1 x + {}^3C_2 x^2 + {}^3C_3 x^3)^4 ({}^2C_1 y + {}^2C_2 y^2)^4$$

which is equivalent 0 coeff. of

y^2, xy, x^2 in

$$(3 + 3x + x^2)^4 (2 + y)^4$$

$$= {}^4C_2 \times 2^2 \times 3^4$$

$$+ {}^4C_2 \times 3^4 \times {}^4C_1 \times 2^3$$

+

$${}^4C_2 \times 3^4 \times 2^4$$

+

$${}^4C_1 \times 3^3 \times 2^4 = 3^4 \left(2^4 + 128 + 96 + \frac{64}{3} \right)$$

$$= 21816$$

Exercise C1

Q1. Seven different coins are to be divided amongst three persons. If no two of the persons receive the same number of coins but each receives atleast one coin & none is left over, then the number of ways in which the division may be made is:

- (A) 420 (B) 630 (C) 710 (D) none

Ans. B

Sol. -

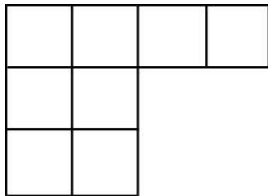
Q2. The number of ways in which a mixed double tennis game can be arranged from amongst 9 married couple if no husband & wife plays in the same game is:

- (A) 756 (B) 3024 (C) 1512 (D) 6048

Ans. C

Sol. -

Q3. The number of ways in which 5 X's can be placed in the squares of the figure so that no row remains empty is:



- (A) 97 (B) 44 (C) 100 (D) 126

Ans. B

Sol. -

Q4. If all the letters of the word "QUEUE" are arranged in all possible manner as they are in a dictionary, then the rank of the word QUEUE is:

- (A) 15th (B) 16th (C) 17th (D) 18th

Ans. C

Sol. -

Q5. The number of integers which lie between 1 and 10^6 and which have the sum of the digits equal to 12 is:

- (A) 8550 (B) 5382 (C) 6062 (D) 8055

Ans. C

Sol. -

Q6. Number of ways in which a pack of 52 playing cards be distributed equally among four players so that each may have the Ace, King, Queen and Jack of the same suit is:

- (A) $\frac{36!}{(9!)^4}$ (B) $\frac{36! \cdot 4!}{(9!)^4}$ (C) $\frac{36!}{(9!)^4 \cdot 4!}$ (D) None

Ans. B

Sol. -

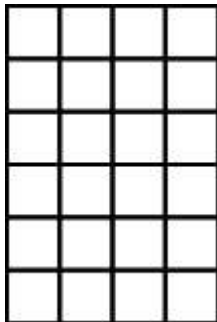
Q7. Number of divisors of the form $4n + 2$ ($n \geq 0$) of the integer 240 is –

- (A) 4 (B) 8 (C) 10 (D) 3

Ans. A

Sol. -

Q8. A rectangle with sides $2m - 1$ and $2n - 1$ is divided into squares of unit length by drawing parallel lines as shown in the diagram, then the number of rectangles possible with odd side lengths is



- (A) $(m + n + 1)^2$ (B) 4^{m+n-1} (C) m^2n^2 (D) $mn(m + 1)(n + 1)$

Ans. C

Sol. -

Q9. If r, s, t are prime numbers and p, q are the positive integers such that their LCM of p, q is $r^2t^4s^2$, then the numbers of ordered pair of (p, q) is

- (A) 252 (B) 254 (C) 225 (D) 224

Ans. C

Sol. -

Q10. There are n identical red balls & m identical green balls. The number of different linear arrangements consisting of " n red balls but not necessarily all the green balls" is ${}^x C_y$ then

- (A) $x = m + n$, $y = m$ (B) $x = m + n + 1$, $y = m$ (C) $x = m + n + 1$, $y = m + 1$
 (D) $x = m + n$, $y = n$

Ans. B

Sol. -

Q11. Let $S = \{1, 2, 3, 4\}$. The total number of unordered pairs of disjoint subsets of S is equal to

- (A) 25 (B) 34 (C) 42 (D) 41

Ans. D

Sol. -

Q12. ${}^{2n}P_n$ is equal to

- (A) $(n + 1)(n + 2) \dots (2n)$ (B) $2^n [1 \cdot 3 \cdot 5 \dots (2n - 1)]$
 (C) $(2) \cdot (6) \cdot (10) \dots (4n - 2)$ (D) $n! ({}^{2n}C_n)$

Ans. D, C, B, A

Sol. -

Q13. In an examination, a candidate is required to pass in all the four subjects he is studying. The number of ways in which he can fail is

- (A) ${}^4P_1 + {}^4P_2 + {}^4P_3 + {}^4P_4$ (B) $4^4 - 1$ (C) $2^4 - 1$ (D) ${}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4$

Ans. D, C

Sol. -

Q14. There are 12 points in a plane of which 5 are collinear. The number of distinct quadrilaterals which can be formed with vertices at these points is:

- (A) $2 \cdot {}^7P_3$ (B) 7P_3 (C) $10 \cdot {}^7C_3$ (D) 420

Ans. D, A

Sol. -

Q15. The number of ways in which 200 different things can be divided into groups of 100 pairs is:

- (A) $\frac{200!}{2^{100}}$ (B) $\left(\frac{101}{2}\right)\left(\frac{102}{2}\right)\left(\frac{103}{2}\right) \dots \left(\frac{200}{2}\right)$ (C) $\frac{200!}{2^{100}(100)!}$
 (D) (1. 3. 5.....199)

Ans. D, C, B

Sol. -

Q16. Number of ways in which 3 numbers in A.P. can be selected from 1, 2, 3, n is :

(A) $\left(\frac{n-1}{2}\right)$ if n is even (B) $\frac{n(n-2)}{4}$ if n is odd (C) $\frac{(n-1)^2}{4}$ if n is odd

(D) $\frac{n(n-2)}{4}$ if n is even

Ans. D, C

Sol. -